



FATIMA JINNAH WOMEN UNIVERSITY

Submitted To:

Engr. Muhammad Shoaib

Submitted By:

Irraj Samreen(027)

Section:

5A

CLOUD COMPUTING

Lab No.15

LAB PROJECT

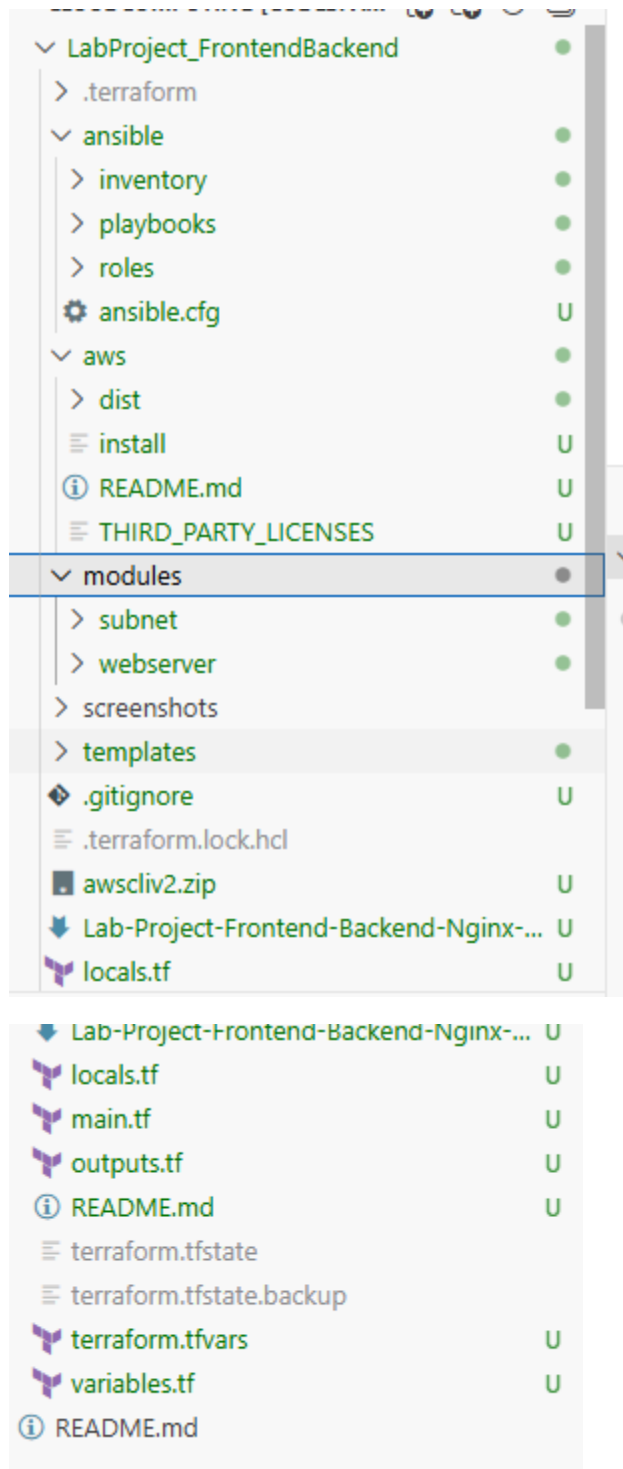
Github Repo Link:

<https://github.com/23-22411-027-max/CloudComputing.git>

Project Overview:

```
@23-22411-027-max →/workspaces/CloudComputing (main) $ ^C
@23-22411-027-max →/workspaces/CloudComputing (main) $ mkdir LabProject_FrontendBackend
cd LabProject_FrontendBackend
@23-22411-027-max →/workspaces/CloudComputing/LabProject_FrontendBackend (main) $ pwd
/workspaces/CloudComputing/LabProject_FrontendBackend
@23-22411-027-max →/workspaces/CloudComputing/LabProject_FrontendBackend (main) $ touch main.tf
variables.tf locals.tf outputs.tf terraform.tfvars .gitignore
@23-22411-027-max →/workspaces/CloudComputing/LabProject_FrontendBackend (main) $
```

Project Structure:



1. Terraform – Networking & Common Settings

[Preview] README.md

.gitignore U X

LabProject_FrontendBackend > .gitignore

```
1  # Terraform
2  .terraform/
3  *.tfstate
4  *.tfstate.backup
5  *.lock.hcl
6
7  # Keys
8  *.pem
9  id_rsa*
10 *.key
11
12 # Ansible
13 ansible/inventory/hosts
14
15 # OS / Editor
16 .vscode/
```

PROBLEMS

OUTPUT

TERMINAL

...

bash - LabProject_FrontendBackend

+

✓

🗑

...

|

🔍

🔍

● @23-22411-027-max → /workspaces/CloudComputing (main) \$ ls

variables.tf variables.tf locals.tf terraform.tfvars.tf main.tf

LabProject_FrontendBackend > variables.tf

```
1  variable "vpc_id" {
2    description = "ID of the existing VPC to use"
3    type        = string
4  }
5
6  variable "subnet_id" {
7    description = "ID of the existing subnet to use"
8    type        = string
9  }
10
11 variable "region" {
12   description = "AWS region"
13   type        = string
14 }
15
16 variable "availability_zone" {
17   description = "Availability Zone"
18   type        = string
19 }
20
21 variable "public_key" {
22   description = "EC2 key pair name"
23   type        = string
24 }
25
26 variable "private_key" {
27   description = "Path to private key file for Ansible"
28   type        = string
```

In:41 Col:1 Span:1

variables.tf variables.tf locals.tf terraform.tfvars.tf main.tf

LabProject_FrontendBackend > locals.tf

```
1  # locals.tf
2  data "http" "my_ip" {
3    url = "https://icanhazip.com"
4  }
5
6  locals {
7    my_ip = "${chomp(data.http.my_ip.response_body)}/32"
8  }
9  |
```

```
.gitignore  variables.tf  locals.tf  terraform.tfvars  main.tf  ×  🔍  📄  .

LabProject_FrontendBackend > main.tf
66 resource "aws_security_group" "sg" {
68   description = "Allow SSH and HTTP"
69   vpc_id      = aws_vpc.main.id
70
71   # SSH from your public IP
72   ingress {
73     from_port = 22
74     to_port   = 22
75     protocol  = "tcp"
76     cidr_blocks = [local.my_ip]
77   }
78
79   # HTTP from anywhere
80   ingress {
81     from_port = 80
82   }
83 }

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  ...  📄  .

▼ TERMINAL  bash - LabProject_FrontendBackend  +  📄  🗑️  ...  >
@23-22411-027-max → /workspaces/CloudComputing/LabProject_FrontendBackend (main) $ terraform
init
terraform apply -auto-approve
aws_route_table_association.public_subnet: Creating...
aws_route_table_association.public_subnet: Creation complete after 1s [id=rtbassoc-0acab2803
0fe56b11]

Apply complete! Resources: 6 added, 0 changed, 0 destroyed.
@23-22411-027-max → /workspaces/CloudComputing/LabProject_FrontendBackend (main) $
```

2. Terraform – Frontend & Backend EC2 Instances

ignore U variables.tf U locals.tf U terraform.tfvars U main.tf U X

LabProject_FrontendBackend > main.tf

```
65 resource "aws_security_group" "sg" {
96   }
97 }
98
99 # Frontend EC2
100
101 resource "aws_instance" "frontend" {
102   ami           = "ami-00db3746fb4246409"
103   instance_type = var.instance_type
104   subnet_id     = aws_subnet.public.id
105   key_name      = var.public_key
106   security_groups = [aws_security_group.sg.name]
107
108   tags = {
109     Name = "${var.env_prefix}-frontend"
110   }
111 }
112
113
114 # Backend EC2s
115
116 resource "aws_instance" "backend" {
117   count = 3
```

PROBLEMS OUTPUT DEPLOY CONSOLE TERMINAL

LabProject_FrontendBackend > main.tf

```
114 # Backend EC2s
115
116 resource "aws_instance" "backend" {
117     count          = 3
118     ami            = "ami-00db3746fb4246409"
119
120     instance_type = var.instance_type
121     subnet_id     = aws_subnet.public.id
122     key_name       = var.public_key
123     security_groups = [aws_security_group.sg.name]
124
125     tags = {
126         Name = "${var.env_prefix}-backend-${count.index + 1}"
127     }
128 }
129
130
131 # Outputs
132
133 output "frontend_public_ip" {
134     value = aws_instance.frontend.public_ip
135 }
```

3. Ansible – Global Config & Inventory

```
TERMINAL nano - LabProject_FrontendBackend + v [] [W] ...
GNU nano 7.2 ansible/ansible.cfg *
[defaults]
host_key_checking = False
interpreter_python = /usr/bin/python3
```

```
GNU nano 7.2 ansible/inventory/hosts *
[frontend]
frontend_ip_here

[backends]
backend1_ip_here
backend2_ip_here
backend3_ip_here

[all:vars]
ansible_user=ec2-user

@23-22411-027-max →/workspaces/CloudComputing/LabProject_FrontendBackend (main) $ terrafo
init
terraform validate

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
Success! The configuration is valid.
```

4. Ansible Roles – Backend HTTPD Role (backend/)

variables.tf U locals.tf U ! main.yml U X terraform.tfvars U main.tf

LabProject_FrontendBackend > ansible > roles > backend > tasks > ! main.yml

```
1 - name: Install httpd
2   yum:
3     name: httpd
4     state: present
5     update_cache: yes
6
7 - name: Enable and start httpd
8   service:
9     name: httpd
10    state: started
11    enabled: true
12
13 - name: Deploy backend index page
14   template:
15     src: backend_index.html.j2
16     dest: /var/www/html/index.html
17     owner: apache
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

variables.tf U locals.tf U backend_index.html.j2 U X terraform.tfvars U

LabProject_FrontendBackend > ansible > roles > backend > templates > backend_index.html.j2

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Backend {{ inventory_hostname }}</title>
5 </head>
6 <body>
7   <h1>Backend server: {{ inventory_hostname }}</h1>
8   <p>Private IP: {{ ansible_default_ipv4.address | default('unknown') }}</p>
9 </body>
10 </html>
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

▼ TERMINAL

@23-22411-027-max → /workspaces/CloudComputing (main) \$ cd ~/workspaces/Cloud bash

```
LabProject_FrontendBackend > ansible > roles > backend > handlers > ! main.yml
1 ---
2 - name: Restart httpd
3   service:
4     name: httpd
5     state: restarted
6
```

5. Ansible Roles – Frontend Nginx Role (frontend/)

```
LabProject_FrontendBackend > ansible > roles > frontend > tasks > ! main.yml
1 ---
2 - name: Install Nginx
3   yum:
4     name: nginx
5     state: present
6     update_cache: yes
7
8 - name: Enable and start Nginx
9   service:
10    name: nginx
11    state: started
12    enabled: true
13
14 - name: Deploy Nginx frontend config
15   template:
16     src: nginx_frontend.conf.j2
17     dest: /etc/nginx/nginx.conf
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL ...

LabProject_FrontendBackend > ansible > roles > frontend > handlers > ! main.yml

```
1  ---
2  - name: Restart nginx
3    service:
4      name: nginx
5      state: restarted
6
```

LabProject_FrontendBackend > ansible > roles > frontend > templates > nginx_frontend.conf.j2

```
1  user nginx;
2  worker_processes auto;
3  pid /run/nginx.pid;
4
5  events {
6      worker_connections 1024;
7  }
8
9  http {
10     upstream backend_servers {
11         server {{ backend1_private_ip }}:80;
12         server {{ backend2_private_ip }}:80;
13         server {{ backup_backend_private_ip }}:80 backup;
14     }
15
16     server {
17         listen 80;
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

TERMINAL

@23-22411-027-max → ~/workspaces/CloudComputing/LabProject FrontendBackend \$ bash

6. Ansible Main Playbook Using Roles

LabProject_FrontendBackend > ansible > playbooks > ! site.yaml

```
1  ---
2  - name: Configure backend HTTPD servers
3    hosts: backends
4    become: true
5    roles:
6      - backend
7
8  - name: Configure frontend Nginx load balancer
9    hosts: frontend
10   become: true
11   vars:
12     backend1_private_ip: "{{ hostvars[groups['backends'][0]].ansible_def
13     backend2_private_ip: "{{ hostvars[groups['backends'][1]].ansible_def
14     backup_backend_private_ip: "{{ hostvars[groups['backends'][2]].ansib
15   roles:
16     - frontend
17
```

7. Terraform–Ansible Integration (Automation)

LabProject_FrontendBackend > main.tf

```
134 # RUN ANSIBLE AUTOMATICALLY (TASK 7)
135
136 resource "null_resource" "run_ansible" {
137   triggers = {
138     frontend_ip = aws_instance.frontend.public_ip
139     backend_ips = join(",", aws_instance.backend[*].public_ip)
140   }
141
142   depends_on = [
143     aws_instance.frontend,
144     aws_instance.backend,
145     local_file.ansible_inventory
146   ]
147
148   provisioner "local-exec" {
149     working_dir = "${path.module}/ansible"
150     # OPTION B: Environment variable tells Ansible where to find the 'ro
151     command = <<-EOT
152     ANSIBLE_ROLES_PATH=./roles ANSIBLE_HOST_KEY_CHECKING=False ansible
153     -i inventory/hosts \
154     playbooks/site.yaml
155     EOT
156   }
```

TERMINAL

@23-22411-027-max → /workspaces/CloudComputing/LabProject_FrontendBackend (main) \$ terraform a
ppl -auto-approve

achable=0 failed=0 skipped=0 rescued=0 ignored=0

null_resource.run_ansible: Creation complete after 55s [id=7868213103277765418]

Apply complete! Resources: 1 added, 0 changed, 1 destroyed.

Outputs:

```
backend_public_ips = [
  "51.112.253.154",
  "3.29.5.116",
  "3.28.201.151",
]
frontend_public_ip = "3.28.191.229"
```

@23-22411-027-max → /workspaces/CloudComputing/LabProject_FrontendBackend (main) \$

⌘ n

In 157 Col 2 Spaces: 2 UTF

Verification & Testing Results

Backend server: 51.112.253.154

Private IP: 10.0.10.203

```
pplly -auto-approve
@23-22411-027-max → /workspaces/CloudComputing/LabProject_FrontendBackend (main) $ ssh -i ~/.ssh/id_ed25519 ec2-user@51.112.253.154
#_
~\ ##### Amazon Linux 2023
~~ \#####\
~~ \###|
~~ \#/ https://aws.amazon.com/linux/amazon-linux-2023
~~ V~' -->
~~~~
~~~~
~/m/'
Last login: Fri Jan 16 17:28:28 2026 from 20.192.21.50
[ec2-user@ip-10-0-10-203 ~]$ sudo systemctl stop httpd
[ec2-user@ip-10-0-10-203 ~]$ 
In 157 Col 2 Spaces: 2 UTF-8
```

← → ↻ ⚠ Not secure 3.28.191.229

Backend server: 3.28.201.151

Private IP: 10.0.10.59

← → ↻ ⚠ Not secure 3.28.191.229

Backend server: 3.29.5.116

Private IP: 10.0.10.119

```
▼ TERMINAL ssh - LabProject_FrontendBackend + ▾ 🗑
@23-22411-027-max → /workspaces/CloudComputing/LabProject_FrontendBackend (main) $ ssh -i ~/
sh/id_ed25519 ec2-user@3.28.191.229
include /etc/nginx/mime.types;

upstream backend_servers {
    server 10.0.10.203:80;
    server 10.0.10.119:80;
    server 10.0.10.59:80 backup;
}

server {
    listen 80;
    location / {
        proxy_pass http://backend_servers;
    }
}
}[ec2-user@ip-10-0-10-4 ~]$
```

```
@23-22411-027-max →/workspaces/CloudComputing/LabProject_FrontendBackend (main) $ ssh -i ~/.  
sh/id_ed25519 ec2-user@3.28.191.229  
}  
}  
EOF  
events {}  
http {  
    server {  
        listen 80;  
        location / {  
            root /usr/share/nginx/html;  
            index index.html;  
        }  
    }  
}  
[ec2-user@ip-10-0-10-4 ~]$ sudo systemctl restart nginx  
[ec2-user@ip-10-0-10-4 ~]$
```

⚠ Not secure 3.28.191.229

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

README.md

```
Site.yaml | terraform.tfvars | main.tf | README.md ^ Q 96 W
① README.md
1 AWS Multi-Tier Infrastructure with Ansible Automation
2 This project demonstrates a high-availability web architecture
  deployed on AWS using Terraform for Infrastructure as Code (IaC) and
  Ansible for automated configuration management. It features a frontend
  Nginx load balancer distributing traffic to three Apache backend
  servers, including a dedicated failover backup server.
3
4 Architecture Overview
5 The infrastructure consists of:
6
7 1 Frontend Server: Running Nginx as a reverse proxy and load balancer.
8
9 3 Backend Servers: Running Apache (httpd) to serve web content.
10
11 Security Group: Configured to allow SSH (port 22) from the
  administrator's IP and HTTP (port 80) from the internet.
12
13 Key Pair: Dynamically deployed for secure SSH access.
14

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL ... | [ ]
```

```
TERMINAL
@23-22411-027-max → /workspaces/CloudComputing/LabProject_FrontendBackend (main) $ terraform c
```

Cleanup

```
@23-22411-027-max → /workspaces/CloudComputing/LabProject_FrontendBackend (main) $ terraform d
estroy -auto-approve
aws_instance.frontend: Still destroying... [id=i-0f9f07f98fd3313ec, 00m40s elapsed]
aws_instance.backend[1]: Still destroying... [id=i-0cb5f70d3dace7085, 00m40s elapsed]
aws_instance.backend[1]: Still destroying... [id=i-0cb5f70d3dace7085, 00m50s elapsed]
aws_instance.frontend: Still destroying... [id=i-0f9f07f98fd3313ec, 00m50s elapsed]
aws_instance.backend[1]: Still destroying... [id=i-0cb5f70d3dace7085, 01m00s elapsed]
aws_instance.frontend: Still destroying... [id=i-0f9f07f98fd3313ec, 01m00s elapsed]
aws_instance.frontend: Destruction complete after 1m0s
aws_instance.backend[1]: Destruction complete after 1m0s
aws_key_pair.deployer_key: Destroying... [id=prod-web-3-3-key]
aws_security_group.sg: Destroying... [id=sg-0b4d4a617ba445334]
aws_key_pair.deployer_key: Destruction complete after 0s
aws_security_group.sg: Destruction complete after 1s

Destroy complete! Resources: 8 destroyed.
@23-22411-027-max → /workspaces/CloudComputing/LabProject_FrontendBackend (main) $
```